Metals

1. Metal chemistry

ID and name: OM1 material_metal_chemistry
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Chemical composition of the metal.

with the two subproperties:

1.1 Chemical composition

ID and name: OM1.1 material_metal_chemistry
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The chemical composition of the metal with additional information.

with the eight subproperties:

1.1.1 Analytical method

ID and name: B4.1 chemistry_method
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The method used to determine the chemical composition.
Allowed values and other constraints: controlled vocabulary

1.1.2 Analysed compound

ID and name: B4.2 chemistry_compound
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The analysed chemical compound (chemical element, oxide, isotope).
Allowed values and other constraints: controlled vocabulary, the actual measured compound must be reported if available (e.g., the isotope for mass spectrometric values rather than the element).

1.1.3 Value

ID and name: B4.3 chemistry_value
Provided by: data provider

Obligation: mandatory
Occurrences: 1–n
Definition: The concentration of the analysed chemical compound.

with the two subproperties:

1.1.3.1 ABSOLUTE VALUE

ID and name: B4.3.1 chemistry_value_absolute
Provided by: data provider
Obligation: recommended
Occurrences: 1–n
Definition: The concentration of the analysed chemical compound.
Allowed values and other constraints: decimal number
Example: 15.3

1.1.3.2 QUALITATIVE VALUE

ID and name: B4.3.2 chemistry_value_relative
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Value of qualitative results and other non-numeric values such as the reporting of detection or quantification limits rather than values.
Allowed values and other constraints: free text
Example: +++, tr, <0.4, bdl = 0.5 wt%

1.1.4 Unit

ID and name: B4.4 chemistry_unit
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The unit in which the concentration of the analysed chemical compound is given.
Allowed values and other constraints: controlled vocabulary

1.1.5 Uncertainty type

ID and name: B4.5 chemistry_uncertainty_type
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The type of analytical uncertainty.
Allowed values and other constraints: controlled vocabulary

1.1.6 Confidence level

ID and name: B4.6 chemistry_uncertainty_sigma
Provided by: data provider
Obligation: recommended

Occurrences: 0–n
Definition: Multiple of standard deviation (sigma) of the reported absolute analytical uncertainty.
Allowed values and other constraints: 1, 2, 3
Example: 2

1.1.7 Uncertainty value

ID and name: B4.7 chemistry_uncertainty_value
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Value of the absolute analytical uncertainty.
Allowed values and other constraints: decimal number
Example: 0.3

1.2 Major elements

ID and name: OM1.2 material_metal_chemistry_major
Provided by: TerraLID system
Obligation: mandatory
Occurrences: 1–n
Definition: Major chemical elements (>1 wt%) in the metal, inferred from the chemical composition.
Allowed values and other constraints: controlled vocabulary

2. Metal corrosion

ID and name: OM2 material_metal_corrosion
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: Information about the corrosion of the metal.

with the two subproperties:

2.1 Extent

ID and name: OM2.1 material_metal_corrosion_extent
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The extent of corrosion affecting the metal.
Allowed values and other constraints: controlled vocabulary

2.2 Details

ID and name: OM2.2 material_metal_corrosion_reason
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Indicators and observations for corrosion.
Allowed values and other constraints: free text
Example: thick green crust with sediment

3. Provenance indicators

ID and name: OM3 material_metal_provenance
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Information about provenance if known from other sources (e.g. stamps).
Allowed values and other constraints: free text
Example: Moulded inscription: Imp(eratoris) Caes(aris) Hadriani Aug(usti) met(alli) Lut(udarensis)